SYSDESIGN REFERENCE

News Feed Capacity Workbook

Traffic, storage, bandwidth, and bottleneck estimation practice.

Rank and deliver personalized items under high fan-out and freshness pressure.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / NEWS-FEED

SYSDESIGN

News Feed Capacity Workbook

CORE IDEA

Rank and deliver personalized items under high fan-out and freshness pressure.

Reference architecture decision chain

01	02	03	04
Fan-Out On Write Vs Read	Ranking	Candidate Stores	Celebrity Users

Decisions to defend

+ Fan-out on write vs read
+ ranking
+ candidate stores
+ celebrity users
+ and cache strategy.

Failure modes to test

+ Hot publishers
+ stale ranking
+ deletion propagation
+ and fan-out backlog.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

News Feed Capacity Workbook

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

REVIEW GATE: TECHNICAL REVIEW. AI-produced draft. A named human must verify technical accuracy, sourcing, freshness, and editorial quality before publication. Reviewed-through date: 2026-07-18.